Roadside Magnetic Sensor System for Vehicle Detection in Urban Environments

Qing Wang, Jianying Zheng, Hao Xu, Bin Xu, and Rong Chen

***Abstract*— Intelligent Transportation Systems (ITS) are widely researched to improve the traffic situation. In ITS, vehicle detection system plays a significant role. At present, vehicle detection is often conducted by inductive loops, which are very expensive and inconvenient to install and maintain. Video camera is another frequently used detector, but it needs high computing power. In order to solve these problems, this paper focuses on the development of a roadside magnetic sensor system for vehicle detection. The device is installed at the side of the road and measures traffic in the adjacent lane (the closest lane to the sensor node). The data are transmitted by the IEEE 802.15.4 communication protocol. A novel adapted threshold state machine algorithm is proposed to detect vehicles. Since false judgments created by large vehicles passing in the nonadjacent lane (the lane next to the closest lane to the sensor node) are frequent in urban environments, a novel feature extracted by fusion of three magnetic sensor signals are proposed to reduce this error. The developed magnetic sensor system is wireless, compact, and cost-effective. The experimental results show that the proposed system achieves high accuracy and is viable in urban environments.**

***Index Terms*— Intelligent transportation systems, magnetic sensors, vehicle detection, wireless sensor network.**

I. Introduction

In urban cities, the traffic situation has become very deteriorating and the traffic congestion is very common, which produces huge economic losses. Intelligent Transportation Systems (ITS) are worldwide researched to alleviate these problems and improve the traffic efficiency [1]. As a part of ITS, real-time vehicle detection plays an important role, since it can gather the fundamental data used in ITS [2]. At present, there are various systems used for vehicle detection, such as inductive loop system, video camera system, microwave radar system, pneumatic tube system and piezoelectric cable system. In these systems, the detectors are first buried under the road surface or equipped above the road. After that, they collect and send data to the traffic management center through wires.

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The shortcomings of these systems are high cost, complicated installation and limited scalability [3]. Since wireless sensor network (WSN) has many advantages, such as low cost, low power consumption, low complexity and self-organizing, the vehicle detection systems based on WSN are proposed to solve these problems [4], [5].

The detection sensor is the key part of a vehicle detection system. The most common detection sensor used in traffic surveillance system is the inductive loop sensor which has the following drawbacks. During the time of installation or maintenance, the inductive loop must be directly buried under the lane, so that the traffic will be interrupted. The cutting seam will soften the pavement and it is easy to damage the road surface. The inductive loop is also easily affected by freezing, saline and busy traffic [2], [6]. Another widely used detection sensor is the video camera. It is a non-intrusive sensor that can be installed above the road so as to avoid damaging the road. Video camera system collects highly accurate traffic information, but requires high computing power, wired power supplies and wired communication. This increases the cost and decreases scalability. The video camera is also easily affected by environmental factors, such as rain, wind, snow and fog [7], [8].

Recently, the magnetic sensor has been used in vehicle detection due to its advantages, such as low cost, small size, high accuracy, and easy installation. The applications of magnetic sensor are often based on WSN. It is a very attractive alternate to the conventional vehicle detection sensors, such as inductive loops and video cameras. Unlike these conventional methods, using magnetic sensor for vehicle detection based on WSN has the following advantages.

1) There is no need to close the lane and cause damage to the road surface. The repair is as convenient as installation.
2) Magnetic sensor detects the vehicles through the changes on earth's magnetic field, so it can't be easily influenced by the climate.
3) There is no reaction to the non-ferromagnetic objects. Therefore, the magnetic sensor can effectively reduce the detection error.
4) The type of vehicle can be roughly determined by the fluctuation range on the magnetic field [9].

Among the researched magnetic sensor systems, there are two main configurations of the sensor nodes. In one configuration, the sensor nodes are buried under the road surface. Whereas, in the other configuration, the sensor nodes are placed at the side of the road. Both of these two configurations have their own advantages and disadvantages. For example, sensors buried under the road surface have stronger signal strength while those placed at the side of the road have easier installation and maintenance. University of California at Berkeley is the forerunner of using magnetic sensor for vehicle detection. Their sensor nodes are designed to deploy in the center of the lane and the installation takes about 10 minutes. During the period of installation and repair, the traffic has to be interrupted [2]. Therefore, the system has the same disadvantage as inductive loop systems. In order to reduce the system cost and the difficulty of installation, most recent researches prefer a more convenient configuration method that deploying the sensor nodes at roadside.

In this paper, "feature" means character calculated by raw data and it can be used in vehicle detection algorithms. "Feature" is extracted from raw data and is used in detection algorithms instead of the raw data. Vehicle detection algorithm based on the extracted feature is of great importance in vehicle detection systems. Most researches use threshold methods to detect vehicles. Because the magnetic field signal is almost stable at a value when there are no vehicles around, and the signal changes a lot when a vehicle passes by. Finally, this signal goes back to the previous stable value when the vehicle leaves. This background magnetic signal value is defined as magnetic baseline. In [10], a fixed threshold state machine algorithm based on a single axis magnetic signal are proposed. In this algorithm, a fixed threshold around the magnetic signal baseline is set first. When a vehicle is passing over the detection area, the magnetic signal fluctuates around the baseline and crosses the threshold. If the duration of this state surpasses a critical value, then the vehicle decision can be generated. Since temperature drift and time drift of the magnetic field can slightly change the magnetic baseline from time to time, the fixed threshold method cannot work efficiently when the drifts occur. In order to solve this problem, in [3] and [11], an adaptive threshold state machine algorithm is used to track the magnetic baseline. In this method, the magnetic baseline keeps updating along with the drifts. To improve the signal-to-noise ratio (SNR), in [12], a feature calculated by cross correlation between the raw signal and a referential signal is obtained first. Then, a simple threshold method is used to detect vesicles. By this way, there is no need to track the magnetic baseline. Another algorithm that can ignore the influence of drifts is used in [13]. In this research, the variance weighting is extracted by 3-axis magnetic signals to reflect the magnetic field fluctuations caused by vehicles. Then a fixed double-threshold state machine algorithm is proposed. When the variance weighting runs over the two thresholds excessively, a vehicle detection decision generates. Similarly, when the variance weighting goes below the two thresholds excessively, a vehicle is considered to leave the detection area. By setting two thresholds, the entry or departure of a vehicle can be guaranteed.

Usually, one magnetic sensor node is set to measure one-lane traffic because of its low sensing distance. However, large vehicles driving in the nonadjacent lane may cause false calls for their strong signal strength. This case is easier to take place in urban cities in which the lane-width is relatively smaller. Until now, little attention has been paid to this circumstance. Unlike the most researched magnetic sensor systems, this paper describes a more convenient system that place the sensor node at the side of the road. The magnetic field angle between y-axis and z-axis is calculated as the vehicle feature and an adaptive threshold state machine algorithm are proposed to realize the vehicle detection. In order to reduce the false detections caused by vehicles driving in the nonadjacent lane, a novel feature extracted by fusion of magnetic sensor array is used to make the system practicable.

The rest of this paper is organized as follows. In Section II, system hardware and sensor node configuration are introduced. In Section III, the vehicle feature calculated by the magnetic signals is first analyzed, and then vehicle detection algorithm is presented. In Section IV, the way to reduce misjudgments in the nonadjacent lane is introduced. In Section V, the experimental results and related analysis are performed. Finally, Section VI draws conclusions and discusses the future works.

II. Hardware Platform and System Configuration

Mother-earth provides us with earth's magnetic field that permeates everything between the south and north magnetic poles. As we know, almost all traffic vehicles have significant amount of ferrous metals in their chassis (iron, steel, nickel, cobalt, etc.). Once a vehicle passing nearby, it will cause disturbances to the magnetic field and a magnetic sensor can be used to capture these disturbances. Appeal to this fact, magnetic sensor is a good candidate for detecting vehicles [14].

The triaxial magnetic sensor Honeywell HMC5883L is chosen in this research for its high sensitivity and low cost. HMC5883L is a fully digital sensor with a 12-bit built-in analog-to-digital converter (ADC). This sensor can achieve 5 milligauss resolution in ± 8 gauss field and each sensor module costs only about 2 dollars [15]. ZigBee is a wireless communication protocol based on IEEE 802.15.4 standard. Since it uses low overhead data transmission and requires low system resources which are vitally important factors for embedded sensor networks. ZigBee is a more suitable communication technology for the wireless sensor networks based ITS applications when compared with other short-range wireless protocols such as Bluetooth and Wi-Fi [16]. Therefore in this paper, the CC2530 wireless transceiver is used for wireless communication. CC2530 is a chip designed by Texas Instruments Inc. (TI) which supports ZigBee protocol. This transceiver works on the 2.4 GHz channel. It has an outdoor communication range of 100 m, required power of 150 mW and its data transmission rate is 250 kb/s [17].

A single printed circuit board (PCB) we designed is shown in Fig. 1. It contains an STM32F103RBT6 microprocessor, three HMC5883L magnetic sensors and a CC2530 wireless transceiver. The magnetic sensors are set at different height and the distance between every two adjacent sensors is 5 cm. When the circuit is running, the microprocessor samples the magnetic sensor output at a rate of 50 Hz. The data is transferred from one PCB to another wirelessly or from a PCB to a laptop through a serial port. Many rest components of this PCB are designed to complete other tasks that have nothing to do with

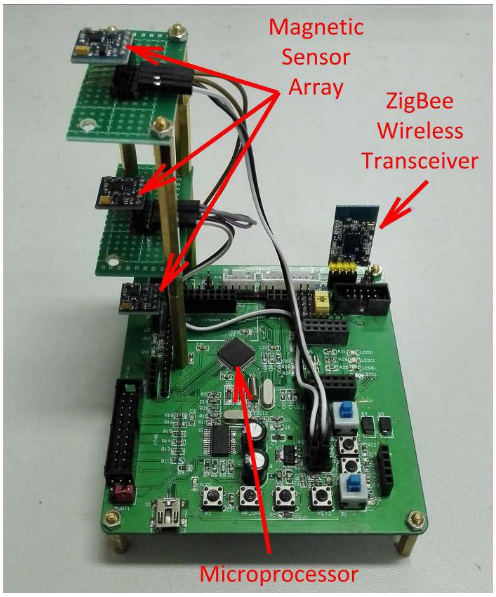

Fig. 1. Developed PCB for sensor node.

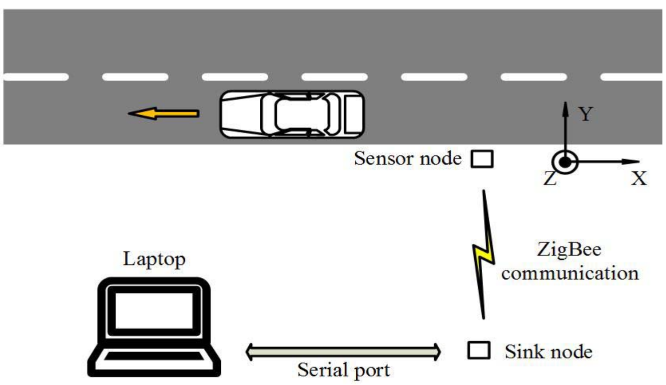

Fig. 2. Sensor node configuration diagram.

this research. In other words, the PCB can be redesigned to have a more compact size if it only operates vehicle detection task.

The PCB works as a sensor node or a sink node of the entire vehicle detection system. The specific way of system configuration is shown in Fig. 2. The sensor node is deployed at the side of the road and the lowest one among the three magnetic sensors is set at a height of 20 cm. The x-axis is along or contrary to the direction of the lane. The y-axis is perpendicular to the direction of the lane, and the z-axis is perpendicular to the road surface and upward. Because the direction of the lane differs in different roads, vehicles may drive in the forward direction or in the reverse direction of x-axis. According to the experiments in [14] and [18], these two circumstances only cause the symmetry of the curve of x, y, and z axes. As a result, the direction of the lane has no influence on the vehicle detection methods used in this research. In our field experiment, the direction of the lane is contrary to the direction of x-axis. This is also shown in Fig. 2. The arrow indicates the direction of the driving

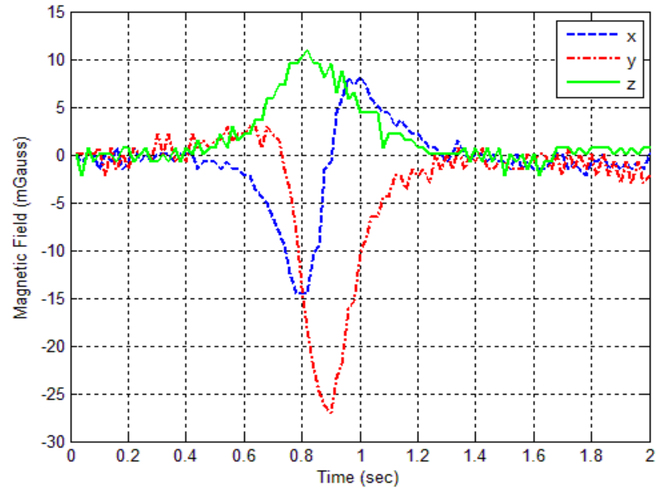

Fig. 3. Magnetic field readings for a sedan.

vehicles. When conducting vehicle detection tasks, the sensor node will measure the magnetic field and transmit data to the sink node by ZigBee communication. Then, the sink node will receive the data and transmit it to the laptop by serial port communication. Finally, the laptop will display and save the traffic measurements. Although the magnetic sensor has three axes, in this paper, only y-axis and z-axis outputs are used in feature extraction and vehicle detection algorithms.

III. Vehicle Detection

A. Signal Filtering and Feature Extraction

As a vector, the magnetic field can be divided into three components in x, y, and z directions. When a vehicle passes by a magnetic sensor, the change of magnetic field caused by the vehicle can be reflected in the output of each sensor axis. Fig. 3 depicts the raw outputs of x, y, and z axis of the sensor at a height of 20 cm when a sedan passing over in the adjacent lane. Since the frequent up-and-down fluctuation of the magnetic signal is not a desirable characteristic for the vehicle detection algorithm, a smoothing filter, which takes a running average of the signal, is used to smooth out the signal [19]. The running average is given by the following equation:

$$B_i(k) = \begin{cases} \dfrac{r_i(k) + r_i(k-1) + \cdots r_i(1)}{k}, & k < M \\ \dfrac{r_i(k) + r_i(k-1) + \cdots r_i(k-M+1)}{M}, & k \geq M \end{cases}$$
$$for\ i \in \{x, y, z\} \quad (1)$$

where $r(k)$ is the raw signal, M is the predefined running average buffer size, and the subscript i represents one of the three axes. The filtered signal of the above sedan is shown in Fig. 4. After filtering, the amplitudes of each signal get a little lower. However, the signals become very smooth and it facilitates the analysis for vehicle signals.

Usually, single axis signal is used as the vehicle detection feature, and z-axis is commonly used because its signal is the most uniform among the three axes. For the magnetic field is a

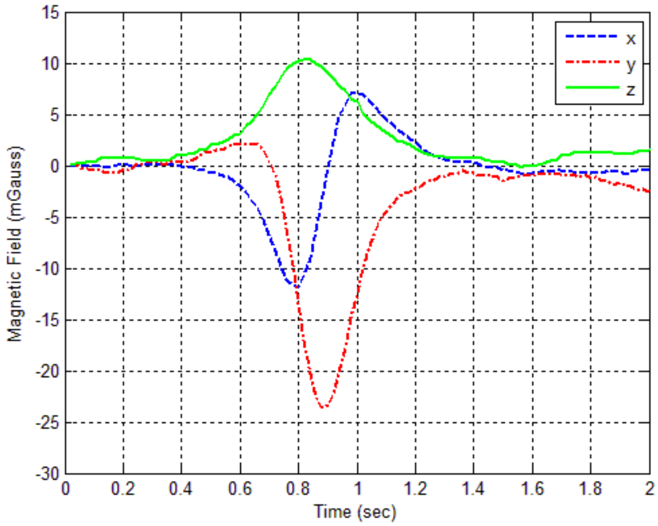

Fig. 4. The filtered signal of the sedan.

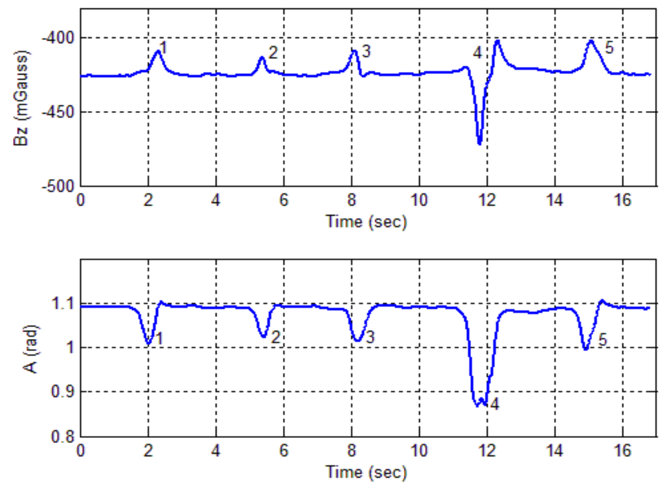

Fig. 5. Comparison of $B_z(k)$ and $A(k)$ signals.

vector, both strength and direction of the magnetic field at the sensor node will change when a vehicle passes by. This paper chooses the vehicle detection feature from the magnetic field direction perspective. We define $A(k)$ as the angle between y-axis and z-axis which can be calculated by the following equation:

$$A(k) = arctan(\frac{B_z(k)}{B_y(k)}) \tag{2}$$

where $B_y(k)$ is the filtered output of y-axis and $B_z(k)$ is the filtered output of z-axis.

Fig. 5 shows the comparison of $B_z(k)$ and $A(k)$ when five vehicles passed by successively. These vehicles are a two-axle truck, a minibus, a pickup, a van and a two-axle truck respectively in order. The vehicle signals of $B_z(k)$ usually only have upward changes. However, observing the forth vehicle signal of $B_z(k)$ which created by a van, it can be seen that this signal has both upward and downward changes. Compared with $B_z(k)$, the whole five vehicle signals of $A(k)$ almost only have downward changes. This behavior makes $A(k)$ very suitable for vehicle detection, because too much vibrations in the signal can easily cause repetitive detections of a single vehicle. Therefore, the feature $A(k)$ we proposed is more stable and more uniform than the commonly used feature $B_z(k)$. Also, compared with the feathers used in other researches, such as cross correlation coefficient and variance weighting, $A(k)$ is computationally simple and has less impact on the real-time running of the sensor node's processor.

B. Vehicle Detection Algorithm

As mentioned above, the angle between y-axis and z-axis is chosen for vehicle detection. Based on this feature, an adaptive threshold state machine algorithm is designed for detecting vehicles in moving traffic. The main reason for using a threshold method instead of other statistical methods is to decrease the calculation amount of the sensor node's processor. As a result, the detection algorithm can efficiently conduct on the sensor node's processor and the detection results can be generated in real time.

1) Adaptive Baseline: Two scenarios will cause uncontrollable drifts in the magnetic signal. One scenario comes from the fact that mother nature will change the earth's magnetic field by small amounts day-to-day and minute-to-minute. Another scenario occurs when temperature on the magnetic sensor changes rapidly. The worst case is on summer cloudy days when the sun comes out of the clouds and suddenly begins baking the housing of the sensor [14]. When the magnetic sensor system works for a long time, these drifts may result of failures if using fixed threshold detection methods.

In order to eliminate the influence of drifts in a long time, an adaptive baseline is first set up to trace the background magnetic readings which can be obtained in the absence of vehicles. Then the adaptive threshold level can be determined by the baseline for the detection state machine. The adaptive baselines for three magnetic sensor axes are calculated by the following equation:

$$L_i(k) = \frac{1}{N}\sum_{j=k-N+1}^{k} B_i(j) \quad for\ i \in \{x, y, z\} \tag{3}$$

Where $L_i(k)$ is the adaptive baseline, $B_i(k)$ is the filtered magnetic signal, and the subscript i represents one of the three axes. The adaptive baseline is only updated on condition that the variance of last successive N readings of each axis is no more than V_0. V_0 represents the variance of successive N readings of each axis when there are no vehicle and other ferrous things around the sensor node. In this way, the baseline of each axis can be accurately updated with no interference. Then, the baseline of $A(k)$ can also be updated by (2).

2) Threshold Selection: A threshold of the variation of $A(k)$ based on the baseline should be set to distinguish whether there is a vehicle passing by or not. By analyzing a quantity of vehicle signals, the threshold level can be empirically selected. Because of the large fluctuations on $A(k)$, almost each vehicle driving in the adjacent lane can be detected by casually choosing a threshold around the baseline. However,

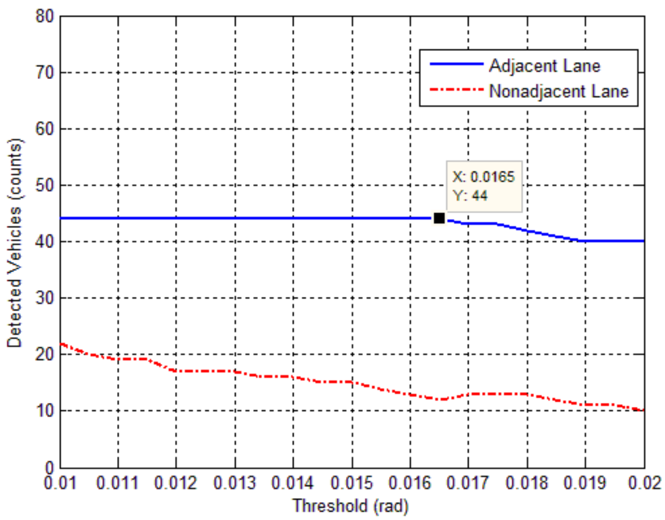

Fig. 6. Vehicle detection results of varying thresholds.

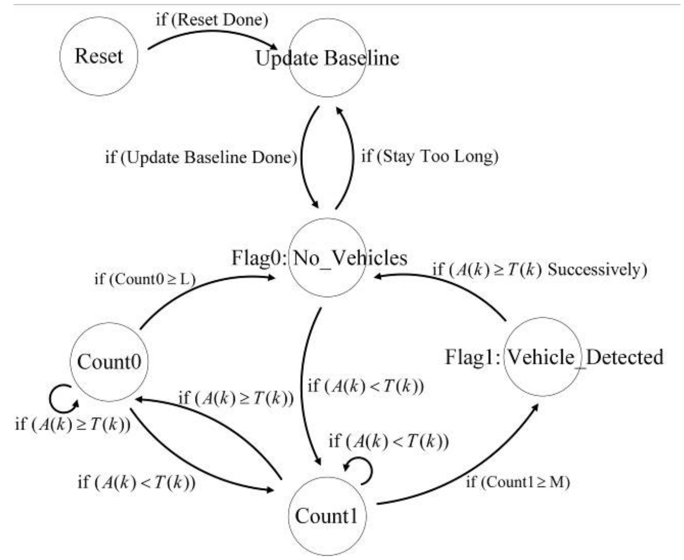

Fig. 7. State machine diagram of vehicle detection algorithm.

many vehicles driving in the nonadjacent lane can also be detected if the threshold is too low. Since the application of this research is to measure traffic in the adjacent lane, vehicle detections in the nonadjacent lane is undesired. As a result, a suitable threshold should be selected to ensure the accuracy in the adjacent lane and reduce the detections in the nonadjacent lane as far as possible at the same time.

A dataset composed of 44 vehicles in the adjacent lane and 75 vehicles in the nonadjacent lane is used for evaluating the threshold. Values range from 0.01 rad to 0.02 rad are selected as the threshold of $A(k)$ to analysis the effect. This range is empirically chosen for the reason that vehicle detection is quite inaccurate out of this range. If the threshold is set lower, more extra detections generate by nonadjacent vehicles or signal noise. If the threshold is set higher, more small vehicles in the adjacent lane cannot be detected. The relationship between detected vehicles and varying threshold is shown in Fig. 6. The blue curve and the red curve represent the detected vehicles in the adjacent lane and in the nonadjacent lane respectively. Both of these two curves have a declining trend, because more and more vehicle cannot pass through the threshold when the threshold grows. It can be seen that the 44 vehicles in the adjacent lane are all correctly detected when the threshold ranges from 0.01 rad to 0.0165 rad. Then the detection rate decreases afterwards. When it comes to vehicles in the nonadjacent lane, the detection rate keeps going down when the threshold increases. As is discussed before, a threshold should be chosen to maximize the detection rate in the adjacent lane and reduce the detections in the nonadjacent lane as far as possible. Therefore, 0.0165 rad is selected as the threshold, because this value keeps a 100% detection rate in the adjacent lane and a relatively low detection rate in the nonadjacent lane.

3) Detection State Machine: The adaptive threshold can be easily calculated based on the adaptive baseline and the selected threshold which mentioned above. In general cases, there should be two thresholds that one is above the baseline and the other is underneath the baseline. However, this paper only cares about the latter because the feature $A(k)$ almost only has downward changes when vehicles passing through the detection area. In this way, the computational complexity of the detection algorithm can be reduced.

$T(k)$ is defined as the threshold underneath the baseline. Then $A(k) < T(k)$ represents that the signal strength exceeds the threshold and this means that a vehicle is entering the detection area. Similarly, $A(k) \geq T(k)$ represents that the signal strength stays around the baseline and does not surpass the threshold. This means that the place near the sensor node is in the absence of vehicles. Therefore in the detection state machine, if the successive sequence of $A(k) < T(k)$ is up to a critical value M, a vehicle is detected. If the successive sequence of $A(k) \geq T(k)$ is up to a critical value L, the detected vehicle is leaving the detection area. The multi-hop detection state machine is shown in Fig. 7. Its main purpose is to convert $A(k)$ to a binary detection flag. The logic flow of the state machine is described as follows.

S1 ("Update Baseline"): Immediately after the system is reset, there must be no vehicles in the detection area. It takes 2 seconds to initialize the baseline by calculating the average outputs of the three axes. Then it will go into state S2. If state S1 is reentered from state S2, the baseline will be updated adaptively.

S2 ("Flag0: No_Vehicles"): After initializing or updating the baseline, it will jump to this state. If the feature $A(k)$ exceeds the adaptive threshold $T(k)$, it will jump to state S3.

S3 ("Count1"): If a vehicle passes by the sensor node, it will produce a successive sequence of $A(k)$ that exceeds $T(k)$. This state is set to track the sequence. Once $A(k)$ is under $T(k)$, it will get into state S4 immediately. If the number of the successive sequence is up to a critical value M, it will go to state S5.

S4 ("Count0"): In this state, if $A(k)$ stays under $T(k)$ continuously and the number of this sequence reaches a critical value L, it will go back to state S2. In order not to miss a potential vehicle detection, it will jump back to the state S3 if $A(k)$ exceeds the threshold $T(k)$ again.

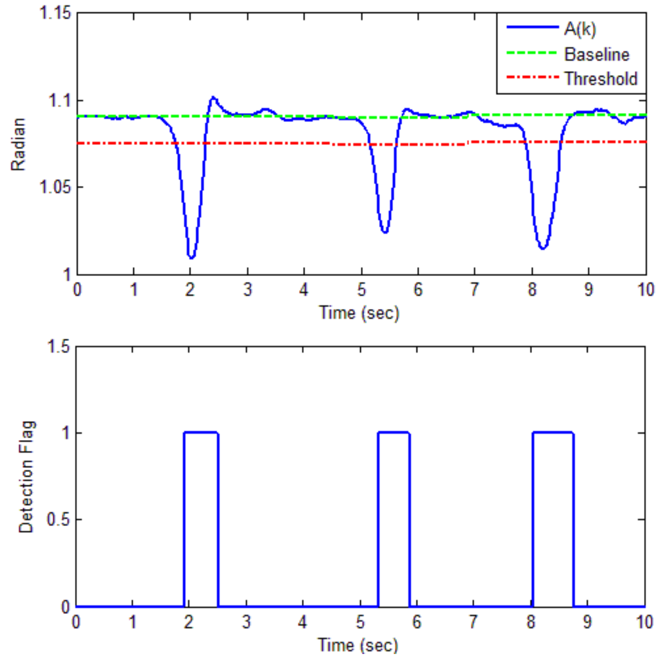

Fig. 8. Vehicle signals and detection flags.

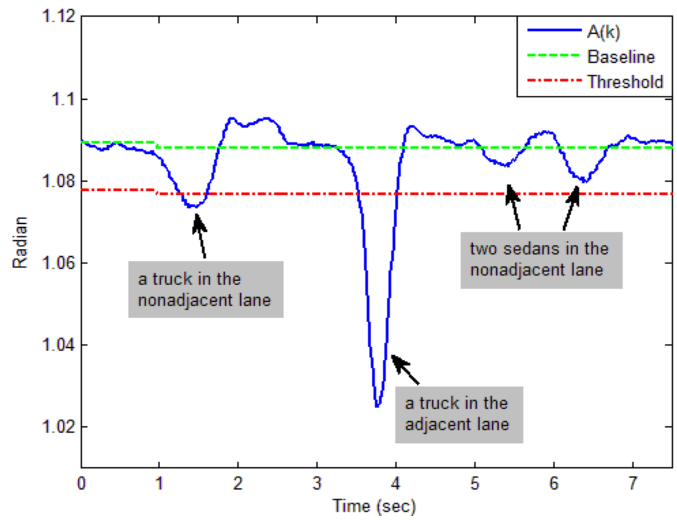

Fig. 9. Misjudgment in the nonadjacent lane.

S5 ("Flag1: Vehicle_Detected"): Getting into this state indicates that a vehicle is successfully detected and staying in state S5 means that the vehicle is still near the sensor node. Consequently, the binary detection flag is set to 1 in the whole S5. If $A(k)$ stays under $T(k)$ successively and the number of this sequence reaches a critical value L, which means the vehicle has left the detection area, it will go back to state S2 and start waiting for a vehicle again.

By running the aforementioned detection state machine, the binary detection flag $D(k)$ can be generated accordingly. Only in state S5, the flag is set to 1. And in other four states, the flag remains at 0. Fig. 8 shows the comparison of vehicle signals and detection flags. In this figure, three vehicles pass over the sensor node successively and cause three obvious downward changes in $A(k)$. Each detection flag changes to 1 shortly after the corresponding vehicle signal. Since the values of M and L are set to 9 and 12 respectively, the slight time delay between a vehicle signal and its detection flag is within 12 measurements. This costs about 0.24s for the sample rate is set to 50 Hz. As a result, the detection can be informed to the processor within 0.24s. Considering that the detection algorithm is simple to compute and has a short time delay, it can be efficiently implemented in the sensor node, and the sensor node can generate vehicle detection report in real-time. Compared with centralized processing systems, this distributed processing way can reduce the data transmission and the power consumption of the system.

IV. Error Reduction to the Nonadjacent Lane

Each vehicle can apparently disturb the Earth's magnetic field at a certain distance, and this distance varies in different vehicles according to how many ferrous metals the vehicle has. Analytical modeling and experimental measurements show that the magnetic field intensity around a vehicle has a relation that approximately varies as $1/x$ with distance, where x is the lateral distance from the vehicle. A Chevy Impala vehicle is tested and it still has a slight magnetic influence at a distance of 6 m [20], [21]. Considering that Chevy Impala is relatively small among vehicles, the influenced distance can be longer if the vehicle is larger than Chevy Impala.

The common lane-width in urban cities is about 3.5 m, which means that the distance between the sensor node and vehicles in the nonadjacent lane is about 4 m in normal driving. Small vehicles such as sedans and Sports Utility Vehicles (SUVs) may have a slight magnetic disturbance on the sensor node and they can be ignored by setting a suitable threshold level. However, large vehicles such as vans, buses and trucks may have a strong magnetic effect that can cause false detections to the system. Fig. 9 shows the four vehicles that go through the direction area successively. In this figure, the second vehicle is the only vehicle which passes by in the adjacent lane and it is correctly detected. The third and the fourth vehicles are two sedans. They are neglected by the detection algorithm, because the signal strength is too low and does not exceed the threshold. The first vehicle is a truck in the nonadjacent lane which still has a strong effect on the sensor node. It runs across the threshold and causes a false detection to the system. This circumstance occurs in urban cities from time to time and the probability of these extra detections in the nonadjacent lane rises when there are more large vehicles driving on the road.

In vehicle detection applications, the magnetic sensor can be embedded in the middle of the lane or be placed at the side of the road. The two configurations both generate detections of vehicles in the nonadjacent lane. In the former configuration, if the road only has two lanes, the sensor can be slightly located toward the outer lane edges to gain a bit of error reduction [14]. But in the latter configuration, setting the sensor node father from the lane cannot help, because small vehicles in the adjacent lane will generate a lower signal and

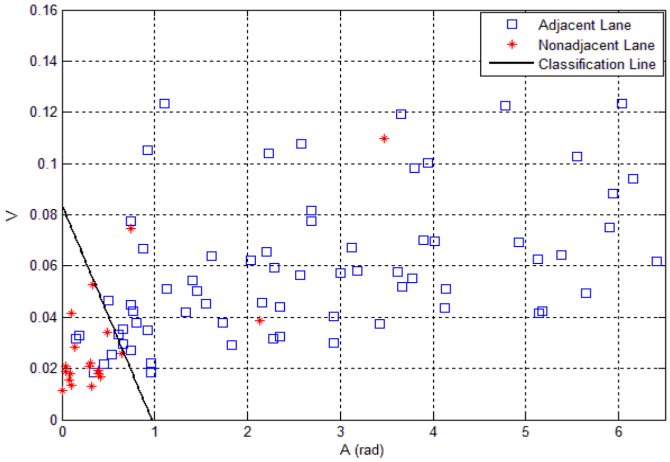

Fig. 10. Use of V and A to reduce error detections in the nonadjacent lane.

be ignored by the detection threshold. The similar solution that increasing the threshold is also a useless way to solve this problem. Although it can reduce the probability of extra detections, a larger threshold will also prevent a part of small vehicles in the adjacent lane from being detected [22]. This is also proven by practical tests in the previous section. In order to relieve this problem, in [23], two magnetic sensor nodes are placed beside the lane edge in different distance. An estimate of the lateral distance of the vehicle is obtained by simply evaluating the ratio of the maximum of the measured magnetic fields between the two sensors. By this way, the detections in the nonadjacent lane can be efficiently reduced. However, in this paper, only one sensor node is used to correct the false detections and the method is proposed as follows.

Since each vehicle consists of many different ferrous parts and the magnetic field intensity produced by the vehicle drops off at a rapid rate with increasing distance from the vehicle. As a result, the lower the distance from the vehicle is, the more drastic fluctuations the magnetic field around has. In order to reflect these fluctuations, three magnetic sensors are implemented at different heights on the sensor node, and the variance of the peak value of each sensor's z-axis signals is calculated when a vehicle passes by. This variance is defined as V in this paper, and the smaller V is, the farther the vehicle is likely to be. In addition, a vehicle that goes through in the nonadjacent lane usually has a much lower peak value $A_{\max}$ compared to a vehicle that passes in the adjacent lane. These two parameters can be combined together to classify whether the vehicle is driving in the adjacent lane or not.

A data set shows that when 115 vehicles passed by in the adjacent lane are correctly detected, there are also undesired 20 vehicles passed by in the nonadjacent being detected. If taking no solutions, these misjudged 20 vehicles will be classified as the vehicles in the adjacent lane, and it leads the total misclassification rate up to 14.8% in these 135 vehicles. Fig. 10 shows the result of applying the proposed method to the data set. A simple linear classification line was set to differentiate the two classes as much as possible. By using this method, finally, only 5 vehicles in the adjacent lane and 3 vehicles in the nonadjacent lane are misclassified. As a result, the total misclassification rate is reduced from 14.8% to 5.9% efficiently, and it makes the number of false detections of the system acceptable.

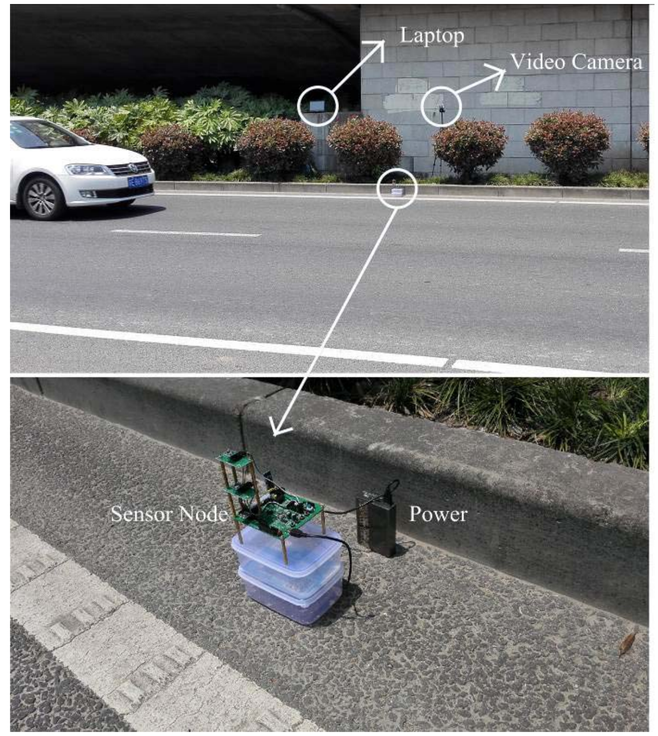

Fig. 11. The scene of on-road experiment.

V. Experimental Results and Analysis

The experimental test was conducted at S227 Road in Xiang-cheng District, Suzhou, China. This road has two lanes with one driving direction from south to north. Since, S227 Road is mainly responsible for shipping the goods, there are more large vehicles driving on this road and the false detection rate of vehicles in the nonadjacent lane can be higher than that on other common city roads. Therefore, this road was typically chosen for the research of false detections. The test was proceeded in a sunny morning and the temperature was about 26 degrees centigrade. The system devices were placed in the left side of moving vehicles. An LTI20-20 Laser velocimeter was used to obtain the speed of vehicles. During the test, the traffic was running normally and the distance between every two vehicles was relatively high. The speed of vehicles ranged from 32 km/h to 67 km/h during the experiments. Fig. 11 shows the test site. The sensor node was placed just next to the lane edge. The sink node and the laptop were deployed several meters away from the road. A video camera was used to shoot local videos, so that the actual vehicles can be counted manually to ensure the accuracy. Since the trial road has two lanes, the system performance in two different scenarios can be both tested. The one scenario is a single-lane road, in which there is no interference caused by the vehicles driving in other lanes. The other scenario is a two-lane road which will generate false detections by vehicles driving in the nonadjacent lane.

TABLE I
RESULT OF THE SINGLE-LANE SCENARIO

Observed Vehicles	Detected Vehicles	Detection Accuracy
81	79	97.5%

TABLE II
DETECTION RATE DISTRIBUTION OF THE VEHICLES IN THE NONADJACENT LANE

Vehicle Type	Observed Vehicles	Detected Vehicles	Detection Rate
Sedan	32	1	3.1%
SUV	8	1	12.5%
Pickup	4	0	0%
Minibus	12	2	16.7%
Bus	2	2	100%
Van	7	7	100%
Two and three axle truck	21	17	91.0%
Four to six axle truck	9	9	100%
Total	95	39	41.1%

TABLE III
RESULT OF THE TWO-LANE SCENARIO

	Total Detected Vehicles	Misclassified Vehicles	Misclassification Rate
Before Processing	118	39	33.1%
After Processing	118	18	15.3%

TABLE IV
COMPARISON WITH SABER TAGHVAEEYAN'S WORK

Paper	This Paper	Saber Taghvaeeyan's Paper
Position of Sensor node	Roadside	Roadside
Vehicle Detection Algorithm	Adaptive Threshold Algorithm	Fixed Threshold Algorithm
Number of Sensor Nodes	1	2
Error Reduction Method	Linear Classification	Support Vector Machine
Vehicle Detection Accuracy in One-lane Scenario	97.5%	99%
Error Reduction Efficiency in Two-lane Scenario	From 33.1% to 15.3%	From 8% to 1%

The experimental test lasted about one hour. After scanning the recorded video, 81 vehicles in the adjacent lane and 95 vehicles in the nonadjacent lane were manually observed. By using the proposed vehicle detection algorithm, 79 out of 81 vehicles in the adjacent lane were correctly detected. Table I shows the result and the detection accuracy is up to 97.5%. It indicates that the detection system can achieve high accuracy in the single-lane scenario. Among the observed 95 vehicles in the nonadjacent lane, 39 vehicles produced a strong signal and were also detected. The probability of generating extra detections in these vehicles is 41.1% and it causes serious excessive detections to the system. If not solving this problem, the system just cannot work in two-lane scenarios normally because only vehicles in the adjacent lane are desired. Table II shows the specific vehicle types and each detection rate in these 95 vehicles. Among the 39 miscounted vehicles in the nonadjacent lane, only 4 vehicles are small vehicles (sedans, SUVs, pickups, and minibuses) and the remaining 35 vehicles are large vehicles (buses, vans, two and three axle trucks, and four to six axle trucks). Therefore, the ratio of large vehicles is 89.7% in the miscounted vehicles. It can be seen that false detections are almost created by large vehicles. If there is a bus, a van or a four to six truck goes through the nonadjacent lane, the system is bound to generate an extra detection. As a result, the system performance are highly relevant to the ratio of large vehicles driving on the monitored road. In other words, the more large vehicles passing by in the nonadjacent lane, the more extra detections generated by the system.

Combining two-lane vehicles together, 118 vehicles are detected. Since 39 out of 118 vehicles are extra detections which caused by nonadjacent vehicles, the misclassification rate is 33.1%. After the processing of error reduction method which mentioned in Section IV, the misclassified vehicles are reduced to 18 and the misclassification rate drops to 15.3%. The result is shown in Table III. It indicates that the system performance has improved a lot after using the proposed method. Compared with the previous data set, the misclassification rate has a little increase because there were more large vehicles driving in the trial road.

To sum up, the experimental results show that the magnetic sensor system can achieve high detection accuracy in single-lane road and the accuracy is up to 97.5%. When measuring two-lane roads, large vehicles passing by in the nonadjacent lane may cause false detections with high possibility. After using the error reduction method, the misclassification rate has dropped from 33.1% to 15.3%. Thus the system can also work in two-lane roads efficiently. This work has some improvements on the relevant researches in the past few years. Compared with [2], [10], and [19], the sensor node is implemented at roadside instead of the center of the road. This way can signifycantly reduce the cost and difficulty of device installation and maintenance. Compared with [3] and [13], false detections generated by vehicles in the nonadjacent lane are considered and a method is proposed to reduce these errors. Regarding the whole work presented in this paper, a highly relevant research is done by Taghvaeeyan and Rajamani [23] at the University of Minnesota. Their research is summarized in [23] in 2014. The detailed comparison between this paper and Saber Taghvaeeyan's work is listed in Table IV. The two researches both implement the sensor node at roadside. To detect vehicles, this paper uses an adaptive threshold algorithm to overcome the drifts of magnetic field rather than a fixed threshold algorithm used in [23]. Both of these two works took the false detections created by vehicles in the nonadjacent lane into consideration. Two sensor nodes were implemented in Saber Taghvaeeyan's research and a Support Vector Machine (SVM) is used to classify the vehicles in different lanes. While in this paper, only one sensor node is implemented and the linear classification method is simply applied. It costs less resources in hardware device. In one-lane field test, the two researches both achieve high detection accuracy. When handling with the problem of false detections in the nonadjacent lane, Saber Taghvaeeyan's work reduced the misclassification rate from 8% to 1%. And the rate was reduced from 33.1% to 15.3% in this paper. It is hard to compare the two data directly because of the different test roads. However, it can be seen that both of the two researches can alleviate the problem of false detections efficiently.

VI. Conclusions and Future Works

In this paper, a portable vehicle detection system based on magnetic sensors is proposed. Unlike the traditional inductive loop systems and video camera systems, this solution is of low installation complexity and low cost. It can be implemented at the side of the road conveniently and does not interfere the traffic during the time of installation and maintenance. In addition, the system transmits data by wireless communication. Thus, it is also scalable. The experimental results show that the system achieves high detection accuracy in the one-lane scenario. The proposed error reduction method can decrease false detections in the two-lane scenario, but it cannot provide accurate vehicle detection and needs more research.

Since only vehicles driving at a normal speed is tested in this research, the system performance may have trouble dealing with congested traffic, in which the vehicle distance is very close to each other and the vehicle magnetic signals may have strong mutual effects. Consequently, vehicle detection methods for low-speed traffic will be researched in the future. Regarding ZigBee network, only one point-to-point link is implemented in this research. This is just a preliminary communication model which is applied in one-node scenario. A more practical network that can be utilized in multi-node scenario will be researched. In addition, vehicle classification is also a research hotspot and will be investigated as well.

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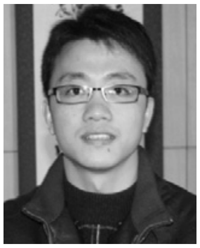

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